## Soru 1


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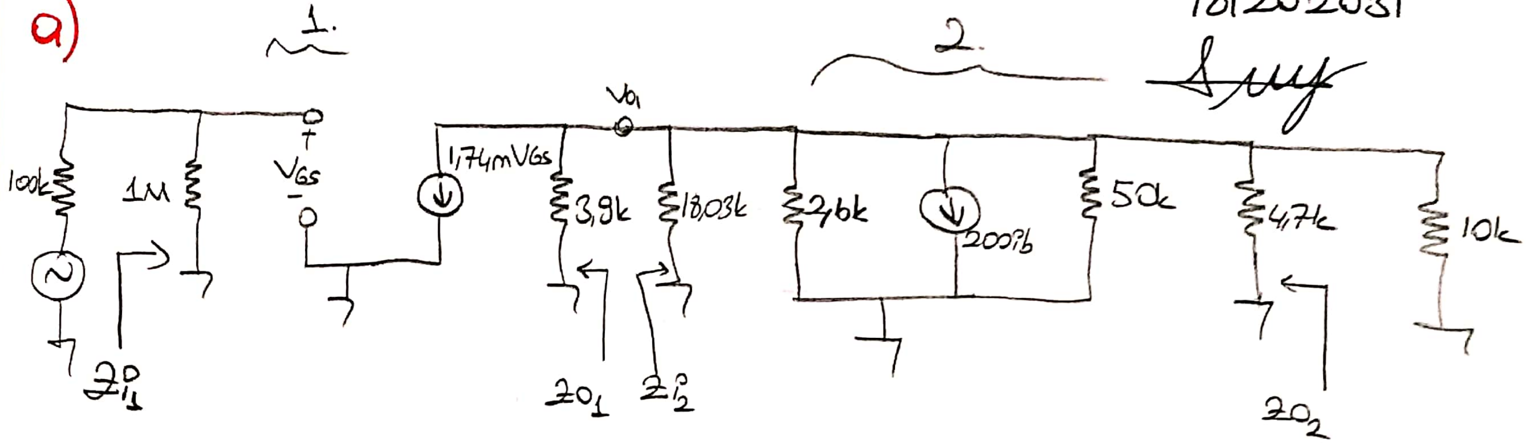
a)




$I_G.1M + V_{GS} + 1{,}5k.I_D = 0 \qquad -1{,}5k I_D = V_{GS} \Rightarrow I_D = 6m\left(1 - \frac{V_{GS}}{V_P}\right)^2$

$\Rightarrow \frac{-V_{GS}}{1{,}5k} = 6m\left(1 + \frac{V_{GS}}{3}\right)^2 \Rightarrow -V_{GS} = 9\left(1 + \frac{2V_{GS}}{3} + \frac{V_{GS}^2}{9}\right) \Rightarrow -V_{GS} = 9 + 6V_{GS} + V_{GS}^2$

$\Rightarrow V_{GS}^2 + 7V_{GS} + 9 = 0$

$V_{GS_1} = \frac{-7+\sqrt{13}}{2} = -1{,}697\,V \qquad V_{GS_2} = \frac{-7-\sqrt{13}}{2} = \cancel{-5{,}302}$

$V_{GS} < V_P$ olacağı için $\boxed{V_{GS} = -1{,}697}$

$g_m = \frac{2I_{DSS}}{|V_P|}\cdot\left(1 - \frac{V_{GS}}{V_P}\right) \qquad g_m = \frac{12m}{3}\cdot\left(1 - \frac{(-1{,}697)}{(-3)}\right) = \boxed{g_m = 1{,}737\ mS}$

$V_B = \frac{15 \cdot 22k}{122k} = 2{,}7V$

2,7V, $I_B$, $V_{BE}$, 1k, $I_E$

$\Rightarrow 2{,}7 - V_{BE} - I_E.1k = 0$

$2{,}7 - 0{,}7 = I_E.1k$

$\boxed{I_E = 2mA} \cong \boxed{I_{CQ} = 2mA}$

$r_e = \frac{26mV}{I_{CQ}} = \frac{26mV}{2mA} = 13\Omega \qquad h_{ie} = \beta.r_e \Rightarrow 200.13 = 2{,}6k$

$\boxed{h_{ie} = 2{,}6k}$

$\boxed{Z_{i_1} = 1M\Omega}$

$Z_{i_2} = 100k // 22k // 2{,}6k$

$\boxed{Z_{i_2} = 2{,}27k\Omega}$

$\boxed{Z_{o_1} = 3{,}9k}$

$Z_{o_2} = 4{,}7k // 50k$

$\boxed{Z_{o_2} = 4{,}29k\Omega}$

$A_{V_1} = -g_m(R_D // R_{i_2}) = -1{,}734(3{,}9k // 2{,}27k) \Rightarrow \boxed{A_{V_1} = -2{,}492}$

$A_{V_2} = \frac{-h_{fe}}{h_{ie}}(R_C // R_y // 50k) = \frac{-200}{2{,}6k}(4{,}7k // 10k // 50k) \Rightarrow \boxed{A_{V_2} = -231{,}143}$

$A_V = A_{V_1}.A_{V_2} \Rightarrow -2{,}492.-231{,}143 = 576{,}008$

$\boxed{A_V = 576{,}008}$

# Soru (1) - Devamı

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b) $C_{C_1} = \dfrac{1}{2\pi \cdot f \cdot (R_g + R_G)} = \dfrac{1}{2\pi \cdot 10 \cdot (1M + 100k)} = 0{,}014\,\mu F$

$C_{C_2} = \dfrac{1}{2\pi f\,(R_D + R_{i_2})} = \dfrac{1}{2\pi 10\,(3{,}9k + 2{,}27k)} = 2{,}579\,\mu F$

$C_{C_3} = \dfrac{1}{2\pi f\,(Z_{o_2} + R_y)} = \dfrac{1}{2\pi \cdot 10\,(4{,}29k + 10k)} = 1{,}113\,\mu F$

c) FET

$C_{in} = C_{gs} + C_{gd}(1 - A_{V_1}) + C_{W_1} \Rightarrow 13{,}476pF = 3pF + 3pF(1 + 2{,}492) + 0$

$R_{in} = 100k \,//\, 1M$

$R_{in} = 90{,}9k\,\Omega$

$f_{C_{in}} = \dfrac{1}{2\pi \cdot 13{,}476 \cdot 10^{-12} \cdot 90{,}9 \cdot 10^{3}} = 129{,}93\,kHz$

$C_{out} = C_{ds} + C_{gd}\left(1 - \dfrac{1}{A_{V_1}}\right) + C_{W_2} \Rightarrow 5{,}2pF = 1pF + 3pF\left(1 + \dfrac{1}{2{,}492}\right) + 0$

$f_{C_{out}} = \dfrac{1}{2\pi \cdot 5{,}2 \cdot 10^{-12} \cdot 1{,}435 \cdot 10^{3}} = 21{,}3\,MHz$

$R_{out} = 3{,}9k \,//\, 2{,}27k$ $\quad R_{out} = 1{,}435k\,\Omega$

BJT

$C_{in} = C_{be} + C_{bc} \cdot (1 - A_{V_2}) + C_{W_1} \Rightarrow 776{,}75pF = 80pF + 3pF(1 + 231{,}15) + 0$ $\quad R_{in} = 1{,}43k\,\Omega$

$f_{c_{in}} = \dfrac{1}{2\pi \cdot 776{,}75 \cdot 10^{-12} \cdot 1{,}435 \cdot 10^{3}} = 143{,}3\,kHz$

$C_{out} = C_{ce} + C_{bc}\left(1 - \dfrac{1}{A_{V_2}}\right) + C_{W_o} \Rightarrow 9{,}01pF = 6pF + 3pF\left(1 + \dfrac{1}{231{,}15}\right) + 0$

$R_{out} = 3{,}9k \,//\, 2{,}27k$

$R_{out} = 1{,}435k\,\Omega$

$f_{c_{out}} = \dfrac{1}{2\pi \cdot 9{,}013 \cdot 10^{-12} \cdot 3{,}005 \cdot 10^{3}} = 5{,}876\,MHz$

$f_{\ddot{u}st_{min}}(F_{c_{in}}, F_{c_{out}}, F_{c_{in_2}}, F_{c_{out_2}}) = \boxed{F_{\ddot{u}st_{min}} = 129{,}93\,kHz}$

# Soru 2

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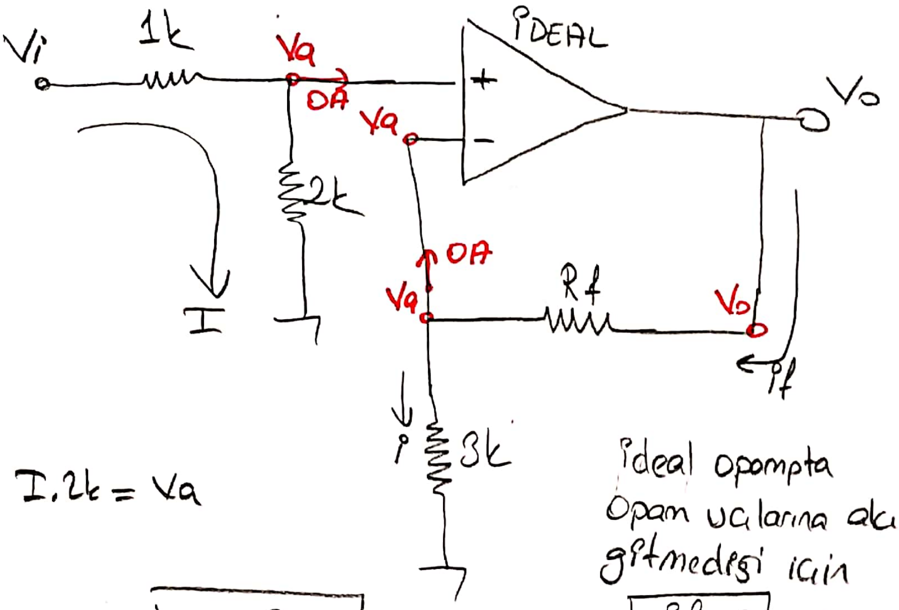


Kazancın 5 olması için Rf

$$\frac{V_o}{V_i} = 5 \text{ için}$$

$\frac{Va}{}$

$$\frac{Vi}{3k} = I \qquad I.2k = Va$$

$$\frac{Vi}{3k}.2k = Va \Rightarrow \boxed{Va = \frac{2}{3}Vi}$$

İdeal opampta Opam uçlarına akım gitmediği için $\boxed{if = i}$ olur

Bu eşitliği kullanırsak;

$$\frac{Vo - Va}{Rf} = \frac{Va - 0}{3k} \Rightarrow 3k.(Vo - Va) = Va.Rf$$

$$\Rightarrow 3k\left(Vo - \frac{2Vi}{3}\right) = \frac{2Vi}{3}.Rf$$

$\Rightarrow \boxed{Vo = 5Vi} \rightarrow$ kazanç

$$\Rightarrow 3k.\left(5Vi - \frac{2Vi}{3}\right) = \frac{2Vi}{3}.Rf$$

$$\Rightarrow 45k\,Vi - 6k\,Vi = 2ViRf$$

$$\Rightarrow 39k\,\cancel{Vi} = 2\cancel{Vi}.Rf$$

$$\frac{39k}{2} = Rf$$

$$\boxed{Rf = 19{,}5k}$$

Soru 3

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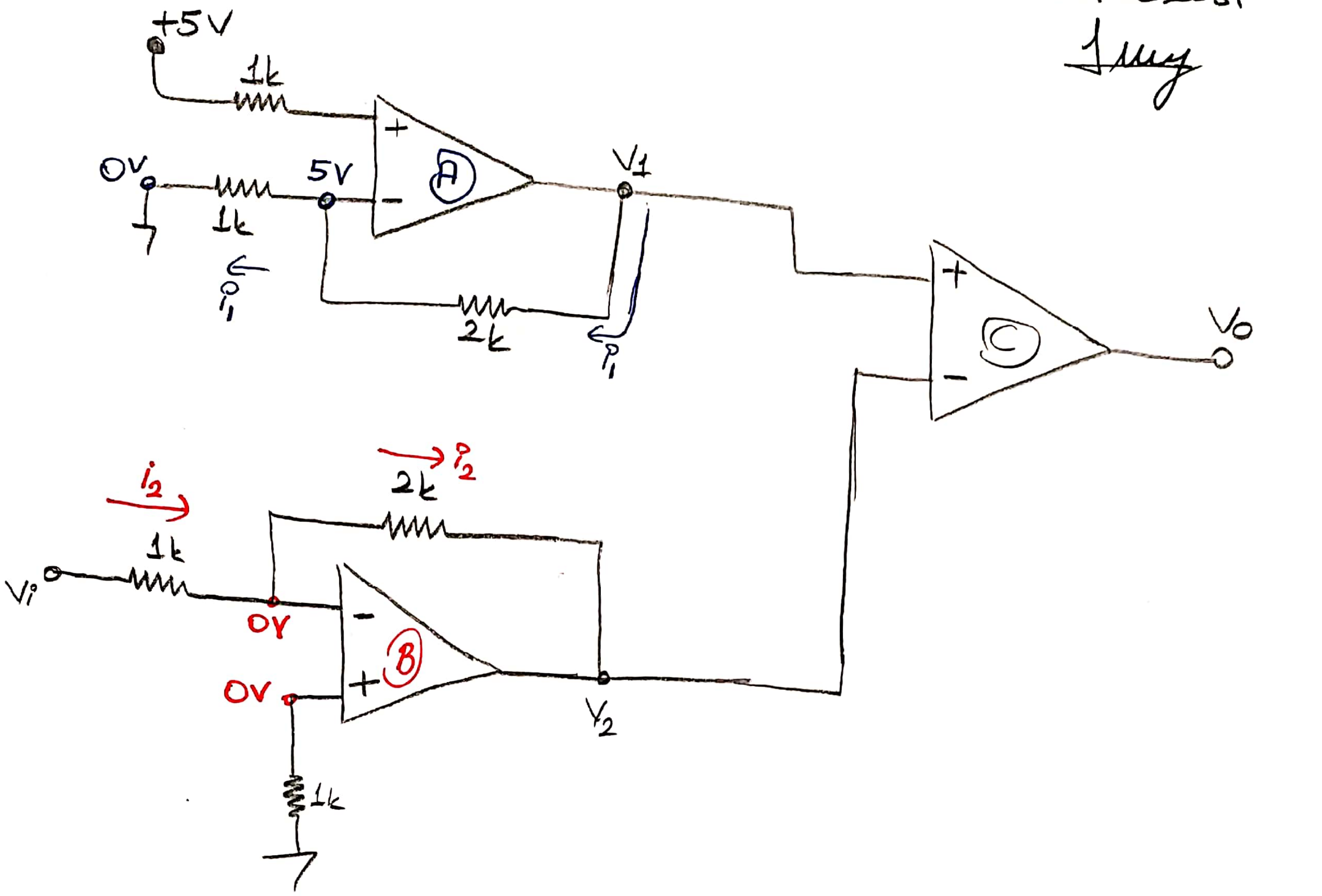


A

$$\frac{5-0}{1k} = \frac{V_1 - 5V}{2k} \Rightarrow 10 = V_1 - 5V \quad \boxed{V_1 = 15V} \text{ ama } V_{cc} = \pm 6V \text{ old. için } \boxed{V_1 = 6V}$$

B

$$\frac{Vi - 0}{1k} = \frac{0 - V_2}{2k} \Rightarrow 2V_i = -V_2 \rightarrow \boxed{V_2 = \pm 2V \text{(sin'in tersi)}}$$

C

$V_1 > V_2$ old. için

$$\boxed{V_o = 6V}$$

# Soru 3 - Devamı

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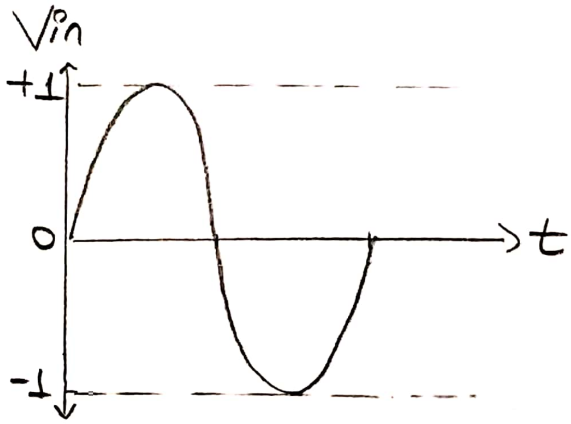


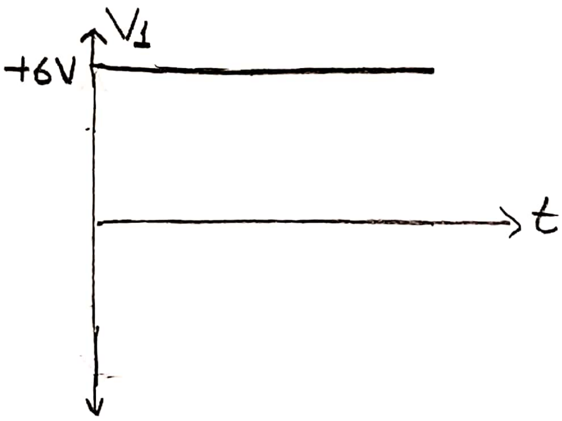


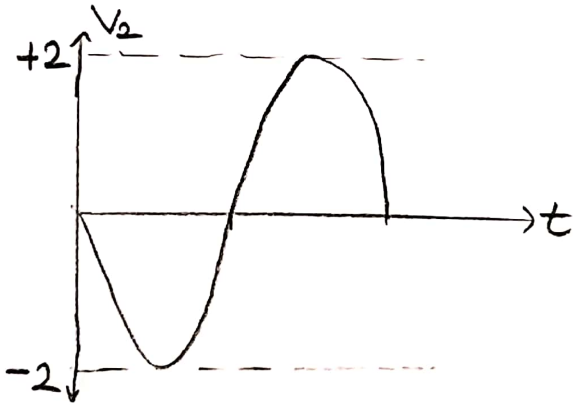


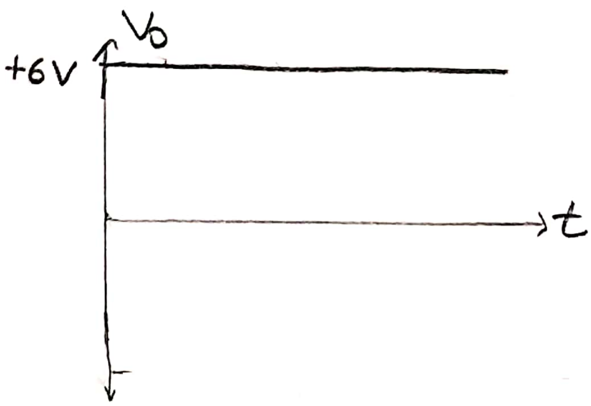

# Soru 4

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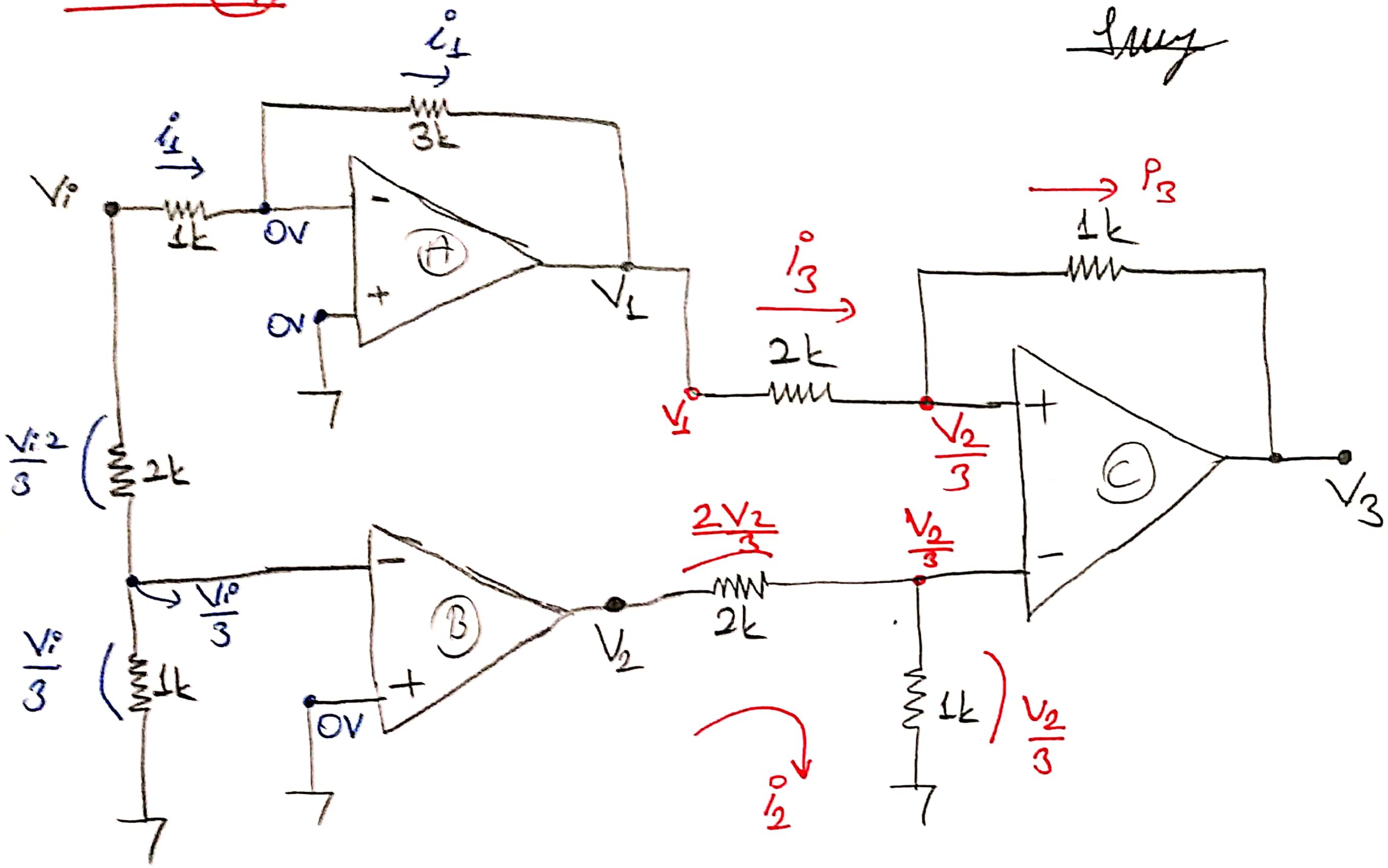


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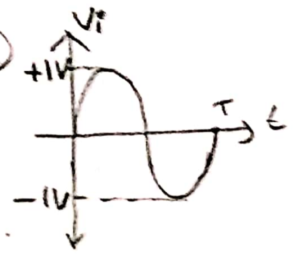


$\boxed{V_{cc} = \pm 12V}$

## A Opampı

$$i_1 = \frac{V_i - 0}{1K} = \frac{0 - V_1}{3K} \Rightarrow \boxed{3V_i = -V_1}$$

## B Opampı

$$\left.\begin{matrix} \frac{V_i}{3} \\ 0V \end{matrix}\right\} \quad V_2 = \begin{bmatrix} \frac{V_i}{3} > 0V & \rightarrow +12V \\ \frac{V_i}{3} < 0V & \rightarrow -12V \\ 0V & \rightarrow 0V \end{bmatrix}$$

## C Opampı

$$i_3 = \frac{V_1 - \frac{V_2}{3}}{2K} = \frac{\frac{V_2}{3} - V_3}{1K}$$

$$i_3 = V_1 - \frac{V_2}{3} = \frac{2}{3}V_2 - 2V_3$$

$$i_3 = 2V_3 = \frac{2V_2}{3} + \frac{V_2}{3} - V_1$$

$$i_3 = \boxed{V_3 = \frac{V_2 - V_1}{2}}$$

# Soru (4) - Devamı

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